Cortical Column Macro-Population Extractor

Optimized Neuron Filtering and 3D Visualization Pre-processing

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1 Project Description

The `extract_macro_populations_debug` function is a highly optimized, intelligent filtering engine designed for large-scale connectomics simulations. Its primary objective is to sift through tens of thousands of neurons within a simulated cortical column to isolate specific macro-populations. This utility is essential for verifying population composition and generating high-fidelity 3D visualizations.

2 Features

- **High-Performance Filtering:** Leverages $O(1)$ dictionary lookups to handle millions of cells efficiently.
- **Smart Routing Logic:** Handles complex biological edge cases, such as the classification of Layer 4 Spiny Stellate cells.
- **NumPy Optimized:** Uses advanced array indexing for near-instantaneous data harvesting.
- **Mutually Exclusive Grouping:** Ensures cells are correctly categorized without overlapping between similar populations.

3 Getting Started

3.1 Prerequisites

Ensure your environment has the following dependencies installed:

- Python 3.8+
- NumPy

3.2 Installation

Clone the repository and include the source in your project path:

```
git clone https://github.com/your-repo/cortical-analysis.git
cd cortical-analysis
```

4 Function Logic Breakdown

4.1 1. The Setup (Unpacking Data)

The function initializes by extracting three core datasets from the `connectomics_data` dictionary:

- `mtype_fast_lookup`: A pre-computed dictionary for rapid classification.
- `cell_mtypes`: A comprehensive list of morphological types.
- `cell_coords`: A spatial array of (X, Y, Z) coordinates.

4.2 2. Parsing the Request

Input strings (e.g., `L23_exc`) are parsed by splitting at the underscore to isolate the `target_layer` and the `target_group`.

4.3 3. The "Fast Scan" Loop

By utilizing `enumerate` on the global `cell_mtypes` array, the function maintains the original indices while hitting the lookup dictionary. This avoids expensive string operations during the primary iteration.

4.4 4. The Smart Routing Engine

The core logic governs cell inclusion based on:

- **Layer Check:** Immediate rejection if the cell layer does not match the target.
- **Inhibitory Check:** Logic for specifically requested inhibitory populations.
- **The Excitatory Conflict:** To prevent Spiny Stellate (`SS`) cells from being double-counted in both `L4_exc` and `L4_ss`, the engine explicitly kicks `SS` cells out of the general excitatory group if a dedicated `L4_ss` group is requested.

4.5 5. Harvesting the Data

Matched indices are converted to NumPy arrays. Through array slicing, the function returns a clean dictionary containing isolated coordinates and labels, optimized for 3D plotting.

5 Usage Example

```
name_list = ["L23_exc", "L4_ss", "L5_inh"]
extracted_data = extract_macro_populations_debug(name_list, connectomics_data)

# Resulting dictionary structure:
# extracted_data['L23_exc'] = { 'coords': array, 'mtypes': array }
```

6 Contributing

Contributions to optimize the filtering logic further are welcome. Please ensure that any changes maintain the $O(1)$ lookup performance.

7 License

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8 Contact

For technical inquiries or support, please contact the maintainers at research@example.com.